Neural Networks Essentials

Introduction

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Outline

Introduction to Neural Networks

Fundamentals and Basic Models

Multilayer Networks and Learning

Case Studies and Practical Insights

Future Directions and Wrap-up

First things first

Everyone gets an NVIDIA redeem code to obtain a certification with NVIDIA Self-Paced Course Program.

Relevant topics include:

- Data Science
- Deep Learning
- Generative AI / LLMs

Link: https://www.nvidia.com/en-us/training/self-paced-courses/

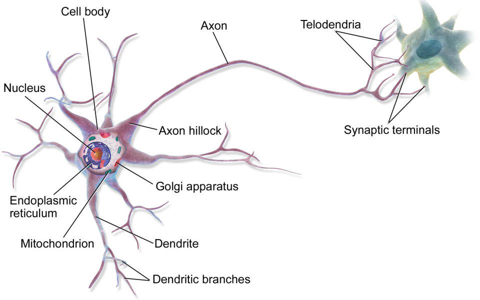
Topics covered in this course

- Introduction to Neural Networks
- Convolutional Neural Networks (CNN)
- Recurrent Neural Networks (RNN) and Sequence Models
- Autoencoders and Representation Learning
- Object Detection and Image Segmentation
- Siamese Networks and Metric Learning
- Self-Supervised and Unsupervised Learning
- Transformers and Attention Mechanisms
- Advanced Topics in Neural Networks
- Practical Applications and Case Studies
- **Any suggestions?**

What are Neural Networks?

- **Definition:** Computational models inspired by the human brain that can learn from data.
- **Motivation:** Solve complex problems in pattern recognition, classification, and decision making.
- **Historical Perspective:** From perceptron (1958) to modern deep learning.
- **Biological Inspiration:** Neurons and synapses in the brain.

Anatomy of multipolar neuron



Overview and Applications

- **Key Applications:**
  - Image and Speech Recognition
  - Natural Language Processing
  - Autonomous systems and Robotics
- **Why Neural Networks?**
  Ability to model nonlinear relationships and learn features directly from data.

The Perceptron Model

- **Concept:** The simplest type of artificial neuron.
- **Mathematical Formulation:**

$$y = \begin{cases} 1 & \text{if } \sum_{i=1}^{n} w_i x_i + b > 0 \\ 0 & \text{otherwise} \end{cases}$$

- **Limitations:** Can only solve linearly separable problems.
- **Improvements:** What can we do?

Activation Functions

- **Role:** Introduce non-linearity into the network.
- **Common Choices:**
  - **Sigmoid:** $\sigma(x) = \frac{1}{1+e^{-x}}$
  - **Tanh:** $\tanh(x)$
  - **ReLU:** $\mathrm{ReLU}(x) = \max(0, x)$
  - **Softmax:** $\mathrm{Softmax}(x_i) = \frac{e^{x_i}}{\sum_{j=1}^{n} e^{x_j}}$
- **Discussion:** Advantages and drawbacks of each.

Structure of a Single Neuron

Biological Inspiration: Neurons in Artificial Neural Networks (ANNs) mimic biological neurons by processing input signals and producing an output.

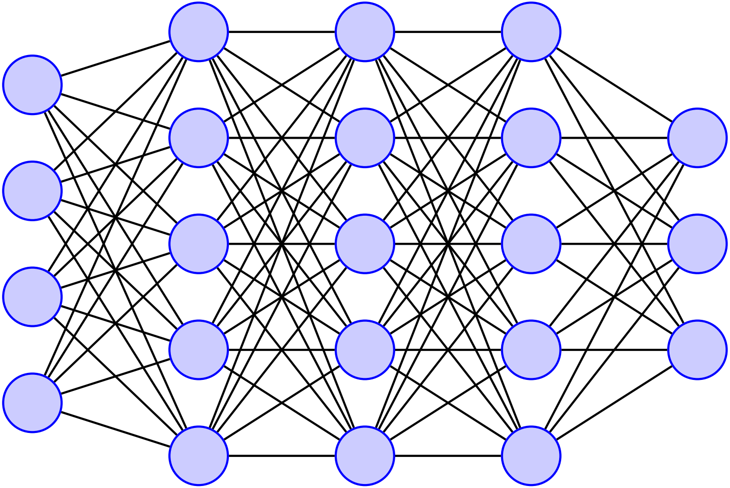
Mathematical Formulation:
A single neuron receives multiple inputs $x_1, x_2, \ldots, x_n$ with associated weights $w_1, w_2, \ldots, w_n$. The neuron computes a weighted sum:

$$z = \sum_{i=1}^{n} w_i x_i + b \tag{1}$$

where b is the bias term.
The output is obtained by applying an activation function σ:

$$a = \sigma(z) \tag{2}$$

Multilayer Neural Networks



Multilayer Neural Networks

- **Architecture:** Input layer, one or more hidden layers, and an output layer.
- **Feedforward Networks:** Data flows in one direction.
- **Universal Approximation Theorem:** Multilayer networks can approximate any function under mild assumptions.

Backpropagation and Training

- **Backpropagation:** Algorithm to update weights by propagating the error backward.
- **Gradient Descent:** Minimizes a loss function by iteratively updating the weights.
- **Mathematical Insight:** Derivation of gradients for weight updates.

Feedforward and Backpropagation

Feedforward:

- Inputs pass through multiple layers.
- Each neuron computes $z = \sum w_i x_i + b$ and applies activation.
- Output layer provides final predictions.

Backpropagation:

- Computes loss between predicted and actual output.
- Uses gradient descent to adjust weights via:

$$w \leftarrow w - \eta \frac{\partial L}{\partial w} \tag{3}$$

- Repeats until convergence.

Loss Functions and Optimization

- **Loss Functions:**
  - Mean Squared Error (MSE) for regression
  - Cross-Entropy Loss for classification
- **Optimization Techniques:**
  - Stochastic Gradient Descent (SGD)
  - Advanced methods: Adam, RMSProp

Regularization and Preventing Overfitting

- **Overfitting:** When the model learns noise instead of the underlying pattern.
- **Regularization Techniques:**
  - L1 and L2 Regularization
  - Dropout: Randomly deactivating neurons during training.
  - Early Stopping

Case Study: Handwritten Digit Recognition

- **Dataset:** MNIST database of handwritten digits.
- **Network Architecture:** Typical MultiLayer Perceptron (MLP) or Convolutional Neural Network (CNN).
- **Key Metrics:** Accuracy, precision, recall.
- **Discussion:** Challenges and improvements.

Tools and Frameworks

- **Popular Frameworks:**
  - TensorFlow, **PyTorch**, Keras
- **Development Environment:** Anaconda, Pip or Colab
- **Live Demonstration:** during the laboratory

Emerging Trends in Neural Networks

- **Deep Learning Evolution:** From traditional MLPs to deep convolutional and recurrent networks.
- **Novel Architectures:** GANs, Transformers, and their applications.
- **Ethical Considerations:** Bias, transparency, and the future of AI.

Summary and Further Resources

- **Key Takeaways:**
  - Neural networks are inspired by the brain but powered by math.
  - Different architectures solve different problems (CNNs, RNNs, Transformers).
  - Deep learning is still evolving – stay curious and keep learning!
- **Additional Resources:**
  - Goodfellow, Ian, et al. **Deep learning**. Vol. 1. No. 2. Cambridge: MIT press, 2016.
  - Bishop, Christopher M., and Nasser M. Nasrabadi. **Pattern recognition and machine learning**. Vol. 4. No. 4. New York: springer, 2006.
  - **NVIDIA Self-Paced Course**

Q & A

Questions?